

A critical evaluation of the Gompertz model in pmsimstats-ng

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1. Motivation and scope

The pmsimstats-ng simulation framework uses a modified Gompertz function as the parametric template for three latent response components: the biological response (BR), the time-varying non-drug component (TV), and the placebo-belief component (PB). The function appears in `R/utilities.R` and in `implementations/{original,original-extended,tidyverse}/R/` collections; it is invoked by the simulation core at every cell of every parameter grid. Because the assumed functional form of the response trajectory is loaded into the simulated outcome before any analysis-model question is asked, the choice of family is not a peripheral implementation detail but a foundational modelling decision. This document evaluates that decision against the relevant biological and biostatistical literature, articulates where the Gompertz form is well-justified for the present setting, identifies where it imposes assumptions the framework does not explicitly defend, and outlines the alternative families that could be used in sensitivity analyses.

2. The Gompertz function

The Gompertz function in its original form is a double-exponential sigmoid

$$y(t) = a \cdot \exp(-b \cdot \exp(-c \cdot t)),$$

where $a > 0$ is an asymptote, $b > 0$ controls displacement of the inflection along the time axis, and $c > 0$ is a rate parameter [gompertz1825nature]. The function was introduced as a demographic model of human mortality and was first repurposed as a growth curve by [winsor1932gompertz], who observed that the sigmoidal shape of biological size-by-time data could be captured parsimoniously by the same three-parameter form. Subsequent work established the Gompertz model as a standard description of tumor growth [laird1964dynamics; nortonSimon1977tumor; norton1988gompertzian], bacterial population dynamics [zwietering1990modeling], and innovation diffusion. A recent review by [tjorve2017gompertz] catalogues parameterisations and emphasises the non-orthogonality of b and c , which complicates identification when both are estimated from data.

In pharmacodynamic modelling, the dominant family is the Hill or Emax model, $y = E_{\max} \cdot t^n / (EC_{50}^n + t^n)$ [holfordSheiner1981; sheiner1989clinical; holford2017pharmacodynamic], which differs from the Gompertz in two respects relevant here: the Hill model is symmetric about its inflection (Gompertz is not), and the Hill model has a closed-form sigmoid centroid at EC_{50} , which has a direct pharmacological interpretation (half-maximal effect concentration). The Gompertz inflection occurs at $t = \ln(b)/c$ and has no comparable interpretive anchor in the drug-response literature.

3. The modified form used in pmsimstats-ng

R/utilities.R implements

```
y <- maxr * exp(-disp * exp(-rate * t))
vert_offset <- maxr * exp(-disp * exp(-rate * 0))
y <- y - vert_offset
y <- y * maxr / (maxr - vert_offset)
```

This is the Gompertz function shifted vertically so that $y(0) = 0$ and rescaled so that $y(\infty) \rightarrow \text{maxr}$. The shift plus rescale maps the family { Gompertz with arbitrary $y(0)$ and asymptote maxr } onto the family { Origin-passing curves with asymptote maxr }. The adjustment is convenient for symptom-improvement modelling because clinical outcome scales (CAPS, PCL, HAM-D, BDI) are typically expressed as change-from-baseline so that 0 is the correct value at study entry.

Two cautions about this implementation. First, the rescaling factor $\text{maxr}/(\text{maxr} - \text{vert_offset})$ becomes numerically unstable as `disp` shrinks because `vert_offset` approaches `maxr`. The current code guards against the degenerate `maxr == 0` case but not against a degenerate `disp` close to zero. Second, the term ‘modified Gompertz’ in the microbiology literature [Zwietering1990modeling] denotes a different reparameterisation, in which a lag time λ and a maximum specific growth rate μ_m are explicit parameters; the pmsimstats-ng usage is unrelated to that established modification, and overloading the name in code comments and manuscripts invites confusion.

4. How Hendrickson et al. (2020) use it

[HendricksonOptimizing2020] use the modified Gompertz to encode three components of the simulated symptom trajectory. The biological response (BR) is a function of cumulative time on drug; the time-varying non-drug component (TV) is a function of calendar time since study entry; the placebo-belief component (PB) is a function of cumulative time under positive expectancy. Each component receives its own $(\text{maxr}, \text{disp}, \text{rate})$ triple. The default calibration is

Table 1: Default modified Gompertz parameters for the BR, TV, and PB components

Component	maxr	disp	rate (per week)
BR	10.99	5	0.42
TV	10.99	5	0.42
PB	6.51	5	0.35

producing curves that rise from zero at study entry, pass through inflection near $t = \ln(5)/0.42 \approx 3.8$ weeks for BR and TV, and approach their asymptotes by approximately week 12. The calibration target is the prazosin-versus-placebo CAPS-difference trajectory in the Raskind active-duty cohort [RaskindTrialPrazosin2013; RaskindReductionNightmares2003; RaskindParallelGroup2007], reproduced as the “Figure 4” reference in the original code base. The BR-versus-TV separation in the Hendrickson model is an architectural choice rather than a mechanism testable from the published data: the Raskind trials report only the total CAPS difference, not a decomposition into biological-response and non-drug temporal components, so the BR and TV maxima are calibrated jointly to a common observed target. This identifiability constraint is acknowledged implicitly in the pmsimstats-ng default parameter sheets but is not, to our knowledge, evaluated empirically in the published literature.

5. Strengths in the symptom-trajectory setting

Sigmoidal shape. Drug-induced symptom improvement in chronic neurobehavioral conditions is widely observed to be sigmoidal rather than linear: there is a delay before measurable benefit, followed by a phase of more rapid improvement, followed by a plateau as the underlying pathophysiology stabilises [Holford2017pharmacodynamic]. A monotone three-parameter sigmoid is therefore a defensible parametric template for the mean structure of such trajectories, and the Gompertz is one of several families that can produce this shape.

Parsimony. Three parameters per component is the minimum needed to represent magnitude, timing, and rate, which is attractive for a simulation framework that must enumerate many parameter grids without combinatorial explosion. Richer families (four-parameter logistic, Weibull cumulative, splines) introduce identifiability problems when the simulation is calibrated to a single observed trajectory.

Computational tractability. The Gompertz is closed-form, infinitely differentiable, and free of the pole and near-zero behaviours that plague the Hill function at $t = 0$. The double-exponential form is numerically stable across the parameter ranges used in the framework.

Established precedent in adjacent fields. The Gompertz has been used for decades in tumor-growth modelling [Laird1964dynamics; NortonSimon1977tumor; Norton1988gompertzian] and bacterial-growth modelling [Zwietering1990modeling], giving the family a level of biological face validity that less commonly used sigmoids do not enjoy.

6. Limitations and concerns

Asymmetry that may not match the data. The Gompertz is more strongly skewed than its biological cousins (logistic, Hill, Weibull cumulative). Its inflection occurs at $1/e$ of the asymptote (approximately 37 percent), rather than at 50 percent as in the logistic. Whether prazosin-induced CAPS improvement in PTSD truly accelerates earlier than it decelerates is an empirical question; the published Raskind trajectories [raskindParallelGroup2018] are not granular enough to settle the question of skewness, and the Hendrickson framework does not report sensitivity to the choice of sigmoid family.

Monotonicity. The Gompertz cannot accommodate response followed by relapse, which is documented in PTSD pharmacotherapy [steckler2007ptsdpharma; hoskins2015ptsd] and chronic depression. Patients who improve early in treatment and then lose benefit at week 8-10 cannot be simulated under any (`maxr`, `disp`, `rate`) choice; the curve is monotone non-decreasing by construction. For a simulation framework whose downstream analyses include the OL+BDC design (open-label titration followed by blinded discontinuation), the inability to simulate within-treatment relapse is a meaningful restriction. The `pmsimstats-ng` framework currently models relapse only as off-drug carryover decay, which conflates two distinct biological mechanisms (treatment escape during continued dosing, versus pharmacological washout after discontinuation).

Plateau is fixed. The asymptote `maxr` is a population-level parameter; there is no participant-level random effect on the plateau. Heterogeneity of treatment effect at the asymptote, which is the empirical phenomenon that motivates predictive-biomarker work in the first place, is grafted onto the Gompertz only via the biomarker-treatment interaction at the architecture layer (Architecture B differential correlation, or Architecture A mean-moderation), not as a property of the Gompertz family. The implicit assumption is that all participants approach the same ceiling at different rates rather than reaching different ceilings; this is a strong assumption that is not, to our knowledge, defended empirically.

Non-orthogonal parameters. The displacement parameter b (`disp`) and the rate parameter c (`rate`) are strongly correlated in the Gompertz likelihood surface [tjorve2017gompertz]. In a calibration setting this is a nuisance; in a simulation setting this manifests as parameter-grid axes that are not really independent. A grid that varies `disp` while holding `rate` fixed is also implicitly varying the inflection time and the early-trajectory slope, which may be undesirable for sensitivity analysis. The `pmsimstats-ng` parameter-sensitivity sweeps in `analysis/scripts/carryover-sensitivity/` and `analysis/scripts/nof1-design-sensitivity/` do not currently isolate these geometric features of the Gompertz curve.

Vertical-offset rescaling can amplify noise. The $y - y(0)$ shift plus rescale produces a curve whose value at any given t is sensitive to small changes in `disp`

because both `vert_offset` and the rescale factor depend on it. The function is well-behaved in the parameter regime used by [hendricksonOptimizing2020] but the framework does not currently flag the regions where the rescale factor diverges.

Departure from established pharmacodynamic practice. The dominant pharmacodynamic family is the Emax/Hill model [holfordSheiner1981; sheiner1989clinical; holford2017pharmacodynamic]. Reviewers familiar with the PK-PD tradition may ask why a Gompertz was preferred to an Emax model whose half-maximal-time parameter has a direct interpretation. The pmsimstats-ng default calibration does not currently report a sensitivity analysis under an Emax alternative, and the published Hendrickson paper does not defend the Gompertz choice over a Hill or Emax form on biological grounds.

Misspecification relative to the analysis model. The analysis model in pmsimstats-ng is a linear mixed-effects model (`nlme::lme(Sx ~ bm + t + Dbc + bm:Dbc, ...)`) with a continuous exposure-decayed predictor `Dbc`. Under the Gompertz DGP, the true mean structure is a sum of three sigmoidal trajectories; the analysis fits this with a linear-in-time term plus a binary-or-decayed-exposure term. The mismatch is intentional, and the simulation studies the consequences for power and Type I error, but the mismatch does mean that the simulated power estimates are conditioned on a specific functional form of the DGP that is itself a modelling choice rather than an established fact.

6b. Implications for the biomarker-treatment interaction test

The founding pmsimstats question is the test of the biomarker-treatment interaction in the mixed-effects model $Sx \sim bm + t + Dbc + bm : Dbc + (1|ptID)$, with $H_0 : \beta_{bm:Dbc} = 0$ [hendricksonOptimizing2020]. The Gompertz-induced response trajectory enters the simulated data through the BR component mean. The seven limitations enumerated in §6 do not contribute equally to the validity of this interaction test. Ranking them by their impact on the bias, power, and coverage of $\hat{\beta}_{bm:Dbc}$:

Most critical for the interaction test.

1. **Fixed plateau (no random effect on `maxr`).** Architecture B encodes the interaction signal in the second-moment BM-BR correlation; Architecture A encodes it in the first-moment mean shift. Both architectures presuppose that between-participant heterogeneity in the biomarker-driven treatment effect manifests in the *speed* of approach to a common ceiling (timing), not in the height of the ceiling itself (magnitude). This is a strong restriction that the biomarker-stratification literature implicitly contests: responder-versus-non-responder heterogeneity is typically conceptualised as different asymptotic benefit, not different approach rates

[@kentPATH2020; @rothwell2005treating]. If the true biological mechanism is plateau-heterogeneous, the Gompertz DGP under-represents the interaction signal that the analysis model attempts to detect, and the simulated power estimates are conservative (biased downward). A reviewer who accepts the latent-subtype framing of Architecture B will reasonably ask why the response ceiling is treated as a population constant.

2. **Monotonicity (no within-treatment relapse).** The Gompertz cannot generate response-then-relapse trajectories. In real PTSD pharmacotherapy data [@hoskins2015ptsd; @steckler2007ptsdpharma] a non-trivial fraction of participants show early benefit followed by decay, often correlated with comorbid depression and chronic pain status. If the true symptom trajectory has this shape and the simulation does not, the linear-in-time analysis model is being asked to fit a smoother surface than it would encounter in practice. The interaction test is then evaluated under conditions kinder than the application; this biases the simulated power upward (optimistic). Combined with point (1), the directions of bias point opposite ways and partially offset, but neither offset is principled.
3. **Misspecification relative to the linear analysis model.** The analysis model fits $\text{bm} + \text{t} + \text{Dbc} + \text{bm:Dbc}$. The true BR mean is sigmoidal, not linear in t . The linear-in- t term absorbs trajectory curvature into a single slope, and any residual curvature is pushed into the random intercept and the AR(1) residual structure. The interaction term bm:Dbc is the residual quantity that survives this absorption; it is therefore the term most exposed to the choice of sigmoid family. A Hill-shaped DGP (more symmetric inflection) and a Gompertz-shaped DGP (asymmetric inflection at $1/e$ of the asymptote) push different amounts of curvature into the linear approximation and may yield different apparent bm:Dbc standard errors at matched calibration targets. The framework currently does not report sensitivity under an alternative sigmoid family; this is the single most actionable Gompertz-related contribution to interaction-test validity.

Moderate impact on the interaction test.

4. **Asymmetric inflection.** The Gompertz inflection at $t = \ln(b)/c$ corresponds to $1/e \approx 0.37$ of the asymptote, in contrast to the logistic 0.5. Under matched `maxr` and matched inflection time, the early-trajectory slope is steeper for Gompertz than logistic, while the late trajectory is flatter. Because the BM-BR correlation under Architecture B is anchored at on-drug timepoints with `tod > 0`, the interaction signal is concentrated in the period where the BR curve is most rapidly changing. A Gompertz BR therefore weights the interaction signal toward early-trajectory observations more heavily than a logistic BR would, and the simulated power has a different mix of contributions across timepoints. The qualitative conclusions are likely robust; the absolute power numbers are not.

5. **Non-orthogonal `disp` and `rate`.** The displacement and rate parameters are correlated in the Gompertz likelihood surface [tjorve2017gompertz]. For interaction-test simulation, this matters when sensitivity sweeps vary one of these parameters. A sweep that walks `rate` while holding `disp` fixed implicitly co-varies the inflection time and the early slope; the resulting effect on the interaction power is a mixture of two geometric properties of the curve, not a clean single-axis sensitivity. The carryover-sensitivity and nofl-design-sensitivity papers do not currently decompose this. A cleaner reparameterisation (Tjorve’s alternative form, or fixing the inflection time as the primary parameter and deriving `disp` as a function of it) would produce sensitivity sweeps with clearer interpretation.

Low impact on the interaction test.

6. **Vertical-offset rescaling instability.** A numerical concern at extreme parameter values. The default (`disp` = 5, `rate` = 0.42) regime is far from the instability region; the interaction test is not affected at the published parameter settings. A guard in the code is prudent but the issue does not undermine current results.
7. **Naming overlap with `Zwietering`’s modified `Gompertz`.** A documentation concern, not a statistical concern. The interaction test is unaffected.
8. **Departure from `Emax/Hill` pharmacodynamic tradition.** A reviewer-facing concern about defensibility of the choice. The interaction test is unaffected at fixed parameters; it would change under an `Emax` DGP only through the same mechanism as point (3) above.

Summary. The Gompertz-related issues that most warrant attention before the next interaction-test manuscript revision are the fixed plateau (point 1), the monotonicity assumption (point 2), and the absence of a sensitivity sweep over sigmoid family (point 3). Points 4 and 5 are second-order. Points 6 through 8 are housekeeping. A direct sensitivity sweep at fixed (`maxr`, inflection time) across Gompertz, logistic 4PL, and Hill sigmoids would address point 3 and is the highest- value next experiment for the founding interaction-test question.

7. Alternative parametric families to consider

Table 2: Alternative parametric response families with strengths, weaknesses, and references

Family	Strength	Weakness	Reference
Logistic (4PL)	Symmetric inflection at 50% asymptote	Same monotonicity problem as Gompertz	[@richards1959flexible]
Hill / Emax	Closed-form pharmacological interpretation; standard in PK-PD	Pole at $t = 0$ for some forms; less natural for time courses than for dose-response	[@holfordSheiner1981]
Weibull cumulative	Tunable shape parameter, recovers exponential at $k = 1$	Three-parameter and identifiability is similar to Gompertz	[@weibull1951statistical]
Bi-exponential	Two-phase response (rapid plus slow)	Non-monotone rate of approach; estimation can be unstable	[@holford2017pharmacodynamic]
Spline / GAM mean structure	Non-parametric, captures relapse and treatment escape	Loses the parsimony argument; harder to calibrate to a single target trajectory	[@brumback1998]
Latent change score / GMM	Allows participant-level trajectory heterogeneity in shape	Heavy machinery; identifiability requires more timepoints than a typical N-of-1 trial provides	[@muthen2001growth]
Stochastic differential equation	Endogenous treatment-escape and relapse	Loses closed-form simulation; substantially heavier compute	[@oksendalSDE]

8. Recommendations for pmsimstats-ng

1. **Document the Gompertz choice as a modelling assumption, not a mechanism.** The pmsimstats-ng manuscripts (notably the companion to

this document at `analysis/report/01-dgp-mean-moderation-vs-mvn/` and the sensitivity papers at `analysis/report/02-carryover-sensitivity/` and `analysis/report/05-nof1-design-sensitivity/`) should state in the simulation-design section that the Gompertz is chosen for parsimony and tractability, that its sigmoidal shape is defensible but not the only defensible choice, and that the power estimates are conditioned on this functional form.

2. **Add a Gompertz-versus-alternative sensitivity sweep.** A direct test of robustness would replace the Gompertz BR with a logistic 4PL and a Hill-form sigmoid at matched asymptote-and-inflection coordinates and report the resulting power changes. The infrastructure for such a sweep already exists in the `analysis/scripts/{carryover,nof1-design}-sensitivity/` pipelines and would require an additional axis in the cell tibble plus a small change to `simulation-core.R` to dispatch on response family.
3. **Rename `modgompertz` or document the deviation from the Zwietering modified form.** Either the function name should be changed to reflect the actual modification (vertical-offset and rescale to origin) or the in-code roxygen documentation should make the deviation from Zwietering explicit. At present, a reader who recognises ‘modified Gompertz’ from the bacterial-growth literature will encounter a different parameterisation under the same name.
4. **Guard the rescale factor.** Add an explicit check on `(maxr - vert_offset)` for near-zero values and either issue a warning or fall back to the unmodified Gompertz at low `disp`.
5. **Consider participant-level variation in `maxr`.** A random-effects extension in which each simulated participant draws $\text{maxr}_i \sim N(\text{maxr}_0, \tau^2)$ would accommodate plateau heterogeneity without abandoning the Gompertz family. This would also bring the framework into better alignment with the latent-class and finite-mixture formulations developed in the companion paper at `analysis/report/03-latent-class-mixture-application/`.
6. **Acknowledge response-then-relapse as a scope exclusion.** Until the framework supports non-monotone response trajectories, manuscripts should state explicitly that the simulation does not capture mid-treatment loss of benefit, and that the reported power estimates may be optimistic for clinical populations in which such loss is documented.

9. Summary

The Gompertz model is a defensible parametric template for the simulated symptom trajectories in `pmsimstats-ng`. It is parsimonious, numerically tractable, and has a long pedigree in adjacent biological domains. It is also

a strong assumption: it imposes monotone non-decreasing trajectories with a fixed population asymptote and a specific asymmetric inflection that may not match the empirical PTSD or chronic-pain symptom-trajectory literature. The current pmsimstats-ng manuscripts use the Gompertz without a sensitivity comparison against the established Emax / Hill alternative or against logistic and Weibull-cumulative families. We recommend that future revisions either defend the Gompertz choice empirically against a candidate alternative or report a sensitivity sweep that demonstrates the power conclusions are robust to the choice of sigmoid family. The Gompertz alone should not bear the weight of a methodological recommendation about trial design.

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